

September 21st, 2025

Dear Editors,

On behalf of my co-authors and myself, I am pleased to submit our manuscript, “Parallel Evolution of Genes Related to Neurological Function and Muscle Development in Urban Birds” for consideration as a Research Article in *Ornithology*.

Urban environments are powerful natural laboratories for studying rapid evolution under anthropogenic change. While genomic evidence of parallel adaptation across cities exists within species, whether such parallels extend across species remains unclear. **Do urban environments drive selection on similar biological processes in closely related species?** In this study, we use whole-genome resequencing of two congeners with overlapping niches but contrasting levels of urban occupancy—the Northern Cardinal (*Cardinalis cardinalis*) and Pyrrhuloxia (*C. sinuatus*)—to test for genomic responses to urbanization. **We then analyze our findings in a broader comparative context alongside published candidate urban-selected gene lists from Burrowing Owls (*Athene cunicularia*) and Great Tits (*Parus major*).** Our results highlight how urban-driven selection can converge across diverse avian taxa, with implications not only for the ornithological community but also for evolutionary biology more broadly. By situating our work within the expanding body of urban avian genomics, we position Ornithology at the forefront of research linking anthropogenic disturbance to evolutionary process.

No part of the research has been published or is in consideration for publication elsewhere. We as the authors have no competing interests and would not receive any direct financial benefit from the publication of this research. All authors have approved of the submission of this manuscript.

Thank you for your consideration of our manuscript for publication in *Ornithology*, and please let me know if there is anything else that you require in your decision.

Sincerely,

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